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15 NOVEMBER 1961
SUPERSEDING
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15 JANUARY 1960

MILITARY SPECIFICATION

COMPASS: M2

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers one type of instrument known as Compass: M2 which incorporates a compass for azimuth readings and a clinometer for angle of site readings.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-111 — Tape, Paper Gummed (Sealing and Securing).

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-601 — Boxes, Wood, Cleated Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-B-676 — Boxes, Setup, Paperboard.

PPP-P-291 — Paperboard, Wrapping, Cushioning.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-117 — Bags, Interior Packaging.

MIL-F-13926 — Fire ControlMateriel: General Specification Governing the Manufacture and Inspection of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures

FSC 6605

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and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U.S. Military Property.

DRAWINGS

ORDNANCE CORPS

A7578443 — Compass: M2.

PUBLICATIONS

ARMY

TM9–6605–200–12T — Repair Parts List for Compass: M2.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model. At the beginning of production, one compass representative of production shall be made available to the Government to determine if the compass meets all the requirements of this specification.

3.2 Materials. The materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction. The compass shall be manufactured in accordance with Drawing A7578443, "Compass: M2" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure the inherent quality and the performance capabilities of the product prior to final acceptance by the Government.

3.4 General specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL–F–13926:

(a) Order of precedence.
(b) Protective finished surfaces.
(c) Special accuracy.
(d) Finish, painting and protective.
(e) Workmanship.

3.5 Environmental.

3.5.1 *Storage temperatures.* The compass shall be subjected to temperatures of plus 160° and minus 80° F. until thermal equilibrium is reached and shall then be thermally stabilized after return to standard ambient temperature (60° to 90° F.), after which, the compass shall show no evidence of physical failure, such as cement separation, glass breakage, or casting failure and shall subsequently meet all the following requirements of this specification.

3.5.2 *Operating temperatures.* The compass shall meet the torque, repeatability and needle locking requirements while exposed to, and thermally stabilized at, ambient temperatures of plus 150° and minus 40° F.

3.5.3 *Vibration.* There shall be no visible damage, cement or compound separation, or loosening of assembled parts as a result of vibration in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of $\frac{1}{16}$ inch ($\frac{1}{8}$ in. max. excursion) for a period of 5 minutes plus or minus 15 seconds with the compass mounted in its normal operating position, except that the cover shall be closed. Subsequent to vibration, the compass shall meet all the requirements of this specification.

3.5.4 *Shock.* The compass with the cover closed shall be dropped from a height of three feet on a solid wood floor, covered with 4 inches of 40 grit kiln dried sand, once with the body horizontal and once with the body vertical. After these drops, the instrument

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shall show no apparent damage and shall meet all the requirements of this specification.

3.5.5 *Sealing.* The compass shall be externally pressurized to 5 psig and shall not show a pressure drop of more than 0.1 psig for a minimum period of 2 minutes.

3.6 Performance.

3.6.1 *Mirror.* The mirror shall be clamped firmly in place and imbedded in sealing compound at two or more points so that the etched line will remain parallel with the open sight when medium torsional pressure is applied to the mirror with the thumbs.

3.6.2 *Window.* The window shall be firmly seated to resist medium torsional pressure, applied with the thumbs, without displacement.

3.6.3 *Needle locking.* When the cover is fully closed, the needle lift mechanism shall clamp the needle firmly against the window. The cover shall remain closed so that the needle lifting (locking) pin does not loosen the cover and permit needle rotation. When cover is opened, the magnetic needle shall swing freely. These requirements shall also be met at the operating temperatures specified in 3.5.2.

3.6.5 *Alinement.* With the body horizontal, the sight arm and cover open flat, and both sight tips vertical; the sight tips, azimuth index, the azimuth zero graduation, the 3,200 graduation, and the etched line on the mirror shall all be in the same vertical plane within 0.020 inch.

3.6.5 *Accuracy.* When the compass is set up on a suitable table or tripod, with the zero graduation in coincidence with the azimuth index, the needle released and oriented in coincidence with the zero graduation and the direction of magnetic north or equal, the table or tripod shall be rotated 1,600 mils to the right and left and the compass needle shall indicate a reading of 1,600 mils within a tolerance of plus or minus 10 mils.

3.6.6 *Compass.*

3.6.6.1 *Magnetic needle.* The needle shall be balanced at the place of manufacture so that it will be level with azimuth scale within plus or minus 0.03 inch when the compass is placed on a leveled (brass) surface plate and the circular level bubble shall be concentric with the red circle within the width of the line forming the red circle.

3.6.6.2 *Repeatability.* The magnetic needle, when set at any poiint on the azimuth scale, shall return to the same position within 10 mils after the needle has been subjected to an amplitude not greater than 800 mils nor less than 200 mils in both directions when subjected to standard ambient temperature (60° to 90° F.) and at each of the operating temperatures specified in 3.5.2.

3.6.6.3 *Half period.* In areas where the horizontal intensity of the earth's magnetic field is not lower than 0.12 oersteds, the average half period shall be 2.5 seconds or less for a minimum of six successive half periods when the magnetic needle is deflected not greater than 400 mils. The half period is defined as the time interval between successive transits of the end of the needle through its point of equilibrium.

3.6.7 *Clinometer.* With the body held in a vertical position, the line of sight of the open sights horizontal, and the bubble of the tubular bubble centered between the level vial graduations, the clinometer scale index shall indicate zero on the clinometer scale within a tolerance of plus or minus 10 mils. Also, with the line of sight inclined 800 mils up or down, and the bubble centered, the clinometer index shall indicate 800 mils on the clinometer scale within plus or minus 10 mils.

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3.6.8 *Torque.* The torque required to operate the hinges, lever, and screw at standard ambient temperature (60° to 90° F.) shall be within the tolerances specified in table I.

TABLE I. *Torque values in ounce inches*

	Minimum	Maximum
Front and rear sight hinge	5	8
Rear sight holder hinge	20	40
Cover hinge	32	48
Elevation scale—index lever	5	8
Declination adjusting screw	5	8

3.6.8.1 When measured at the extreme operating temperatures specified in 3.5.2, the torque values shall not have changed from the respective values obtained at standard ambient temperatures by more than fifty percent.

3.7 **Identification marking.** Unless otherwise specified, identification marking shall be in accordance with Standard MIL–STD–130.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Contractor inspection.** The supplier shall be responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 *Contractor's quality assurance system.* The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

4.1.1.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 *Government verification.* All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to

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measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Production model.

4.2.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

4.2.1.2 These tests shall be conducted at the contractor's facility or Government approved laboratory using test equipment designed or procured for use on the remaining items.

4.2.1.3 The model shall meet all the requirements of this specification using the inspection method prescribed herein.

4.2.1.4 *Disposition of rejected production model.* Should the production model fail to pass the test for Government acceptance, the contractor shall cease those operations which are causing rejectable material and make the necessary corrections. When corrections have been made, a new production model shall be tested by the contractor under the surveillance of the Government. Operations causing rejections shall not be resumed until a production model has successfully passed all inspections specified.

4.3 Inspection provisions.

4.3.1 *Inspection of components.* All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements effecting concentricity, squareness, parallelism, microfinishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a critical requirement. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.3.1.1 *Sampling inspection.* Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table II.

TABLE II. *Sampling plan*

Lot size inclusive	Sample size	Accept lot	Reject lot
0—25	Use 100% inspection		
26—35	20	0	1
36—50	25	0	1
51—150	30	0	1
151—300	30	1	2
301—over	60	2	3

4.3.1.1.1 *Disposition of rejected lots.* Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. All rejected lots shall be screened for all defective characteristics.

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After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection of the characteristics that caused the lot to be rejected.

4.3.1.2 *Control of materials.* Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.3.2 *Lot formation.* Unless otherwise specified by the contracting officer, or his representatives, inspection lot sizes and lot formation shall be in accordance with Standard MIL–STD–105.

4.3.3 *Acceptance tests.* The classification of defects in table III and IV shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirements, and to reject individual non-conforming items.

4.3.3.1 *Disposition of rejected lots.* Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL–STD–105.

TABLE III. *Classification of defects*

For normal inspection use level L7, and sampling plan table IV–A of Standard MIL–STD–105

AQL value is expressed in percent defective

Major: AQL 1.0%	Requirement paragraph	QA paragraph	Test method
101. Storage temperatures	3.5.1	4.4.1.1	Device, special testing (DST)
102. Operating temperatures	3.5.2	4.4.1.2	DST
103. Vibration	3.5.3	4.4.1.3	DST
104. Shock	3.5.4	4.4.1.4	Manual-Visual
105. Sealing	3.5.5	4.4.1.5	DST

TABLE IV. *Classification of defects*

For normal inspection use level II, and sampling plan table IV–A of Standard MIL–STD–105
For reduced inspection use level I, and sampling plan table IV–A of Standard MIL–STD–105

AQL values are expressed in percent defective

Minor: AQL 4.0%	Requirement paragraph	QA paragraph	Test method
101. Needle locking	3.6.3	4.4.2.3	Visual-Manual
102. Alinement	3.6.4	4.4.2.4	DST
103. Acuracy	3.6.5	4.4.2.5	DST
104. Magnetic needle	3.6.6.1	4.4.2.6.1	Visual
105. Repeatability	3.6.6.2	4.4.2.6.2	Visual
106. Half period	3.6.6.3	4.4.2.6.3	Standard Measuring Equipment (SME)
107. Clinometer	3.6.7	4.4.2.7	Visual
108. Torque	3.6.8	4.4.2.8	SME
109. Identification marking	3.7	4.4.2.9	MIL–STD–130
Minor: AQL 4.0%			
201. Mirror	3.6.1	4.4.2.1	Tactile
202. Window	3.6.2	4.4.2.2	Tactile
203. Workmanship	3.4	MIL–F–13926	Visual

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4.3.4 *Inspection equipment.*

4.3.4.1 *Contractor provisions.* Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration and rehabilitation on a regularly scheduled basis, and shall have avaiable, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment, surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.3.4.2 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.4 Test methods and procedures.

4.4.1 *Environmental.*

4.4.1.1 *Storage temperatures.* The compass shall be exposed to temperatures indicated in 3.5.1 by means of a cold chest and oven capable of maintaining these temperatures. The test shall be in accordance with Procedure II of Specification MIL–F–13926. After exposure, the compass shall be visually examined for evidence of damage indicated in 3.5.1 and shall meet the requirements specified.

4.4.1.2 *Operating temperatures.* The compass shall be subjected to the temperatures indicated in 3.5.2 by means of a chold chest and oven. While exposed at these temperatures, the torque requirements of 3.6.8.1 shall be performed with the aid of a commercial pull scale graduated in increments of ounces. The torque values shall be as specified in 3.6.8.1.

4.4.1.3 *Vibration.* With the cover closed to protect the pivot and jewel of the magnetic needle, the compass shall be vibrated in its normal operating position by means of a vibrating testing device capable of vibrating at the number of cycles, amplitude and time as indicated in 3.5.3. After vibration, the compass shall be visually examined for evidence of damage indicated in 3.5.3.

4.4.1.4 *Shock.* The shock test is a visual-manual check and shall be in accordance with the method and requirements specified in 3.5.4.

4.4.1.5 *Sealing.* The compass shall be subjected to an external pressure indicated in 3.5.5. This is accomplished by means of a special testing device capable of making the pressure requirements specified. The pressure drop and time period shall not exceed the requirements of 3.5.5.

4.4.2 *Performance.*

4.4.2.1 *Mirror.* The mirror test is a visual and tactile examination and shall be in accordance with the procedure described in 3.6.1.

4.4.2.2 *Window.* The window test is a tactile examination and shall be in accordance with the procedure described in 3.6.2.

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4.4.2.3 *Needle locking.* This is a visual-manual test and shall be in accordance with the procedure described in 3.6.3.

4.4.2.4 *Alinement.* The alinement test is accomplished by means of a special testing device consisting of a base with a slide and microscope. The slide is provided with a clamping device and table for holding the compass which can be moved in a straight line under the viewing microscope. The test consists of clamping the compass on the slide of the device and focusing the microscope on each of the six items indicated in 3.6.4 by means of the slide and checking for tolerance specified.

4.4.2.5 *Accuracy.* Accuracy of the azimuth scale is checked by means of the above special testing device. The test is made by mounting the compass on the slide and table of the device so that the needle coincides with the "zero" of the azimuth scale. The table component of the device is then revolved to the 1600-mil position, and then the position of the needle is observed with respect to the 1600-mil mark of the azimuth scale. Any deviation shall not exceed the tolerance of 3.6.5.

4.4.2.6 *Compass.*

4.4.2.6.1 *Magnetic needle.* The magnetic needle test is visual and shall be in accordance with the requirements specified in 3.6.6.1.

4.4.2.6.2 *Repeatability.* Repeatability is a visual test and shall be in accordance with the procedures and the requirements specified in 3.6.6.2.

4.4.2.6.3 *Half period.* The average half period time is recorded by means of a standard stop watch, and if this time exceeds the requirements of 3.6.6.3, an investigation shall be made for interfering influences. In cases where no magnetic influence is apparent, laboratory test equipment shall be used for measuring of the earth's magnetic field in terms of minimum oersteds specified.

4.4.2.7 *Clinometer.* This is a visual check and shall be in accordance with the procedures and requirements specified in 3.6.7.

4.4.2.8 *Torque.* Torque tests for hinged operating components are accomplished by means of a scale indicated in 4.4.1.2 and shall meet the requirements at standard ambient temperatures specified in 3.6.8. When measured at temperatures specified in 3.5.2, the change in torque valves shall not exceed those specified in 3.6.8.1.

4.4.2.9 *Identification marking.* Identification marking shall be visually inspected for conformance to Standard MIL–STD–130.

4.5 Preservation and packaging tests. The test procedures in Specification MIL–P–116 shall be used for the following packaging and preservation tests to determine the minimum acceptable quality.

TABLE V. *Classification of defects*
For normal inspection use level II, and sampling plan table IV–A of Standard MIL–STD–105

AQL value is expressed in percent defective

Major: AQL 1.5%	Requirement paragraph	Test method
101. Improper cleaning (level A)	5.3.1.1 5.3.2.1	MIL–P–116 Visual
102. Improper drying (level A)	5.3.1.2 5.3.2.1	MIL–P–116 Visual
103. Unit packaging, publication and compass w/o carrying case; improper method (level A, C).	5.3.1.3.1 5.3.2.2	MIL–P–116 Visual

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TABLE V. *Classification of defects*—Continued

Major: AQL 1.5%	Requirement paragraph	Test method
104. Wrapping; publication omitted, compass not wrapped as illustrated, not secured (level A, C).	5.3.1.3.1.1 Figure 1	Visual
105. Water-vaporproof bag; unsealed, punctured, improperly sealed, excess air not expelled (level A).	5.3.1.3.1.2 5.3.1.3.2.2	MIL–P–116 Visual
106. Bagged compass not immobilized, publication omitted, paperboard box not conforming to specifications, improper closure, not as specified (level A).	5.3.1.3.2.2	PPP–B–676 Visual
107. Unit packaging; publication and compass w/carrying case; improper method (level A, C).	5.3.1.3.2 5.3.2.2	MIL–P–116 Visual
108. Compass w/carrying case; cover not secured, publication omitted, publication and case not wrapped as illustrated, case with compass not immobilized, box not as specified, closure not as specified (level A, C).	5.3.1.3.2.1 5.3.2.2.2	Visual
109. Intermediate packaging; items and stock number incorrect, paperboard boxes not as specified, not immobilized, not conforming to specification, improper closure, tape not conforming to specification, excess weight (level A, C).	5.3.1.4 5.3.2.3	PPP–B–636 Visual
110. Wrapping and paperboard box for compass w/o carrying case, not as specified, wrapped compass with or without publication not immobilized, box not conforming to specification, closure not as specified (level C).	5.3.2.2.1	PPP–B–566 Visual
111. Intermediate containers not immobilized, shipping container not conforming to specifications (level A, B).	5.4.1 5.4.2	PPP–B–601 PPP–B–621 Visual
112. Intermediate containers with unit package compasses with or w/o carrying cases not immobilized, shipping container not conforming to specifications, excess weight (level C).	5.4.3	PPP–B–636 Standard Measuring Equipment (SME) Visual
113. Shipping container not marked in accordance with standard, markings illegible, incorrect, omitted, serial numbers and fragile markings illegible, incorrect, omitted, not as specified (level A, B, C).	5.5.2	MIL–STD–129 Visual
Minor: AQL 4%		
201. Packing; stock number of items in intermediate packaging incorrect, illegible, omitted (level A, B, C).	5.4	Visual
202. Marking; interior packages not in accordance with standard, serial number and fragile marking not as specified (level A, B, C).	5.5.1	MIL–STD–129 MIL–P–116 Visual
203. Stock number illegible, omitted, incorrect (level A, B, C)	5.5.3	Visual
204. Workmanship	5.7	Visual

TABLE VI. *Classification of defects*

For normal inspection use level L7, and sampling plan table IV–A of Standard MIL–STD–105.

AQL value is expressed in percent defective

Major: AQL 6.5%	Method of unit protection	Test
Heat seal (Level A)	IA–8	MIL–P–116
Quick leak (Level A)	IA–8	MIL–P–116

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4.6 Preproduction pack. The preproduction pack (specified in 5.6) shall be inspected 100 percent by the contractor or his representative, and witnessed by the Government inspector to determine conformance with the requirements of this specification for Level A.

4.6.1 *Preproduction pack failure.* Should the preproduction pack fail to pass the test for Government acceptance, packaging operations will cease until the defective characteristics have been corrected in the process used to produce the pilot pack. When corrections in the process have been made, a new pack shall be completely inspected by the contractor and witnessed by the Government inspector.

4.6.2 *Rough handling tests.* The pack (specified in 5.6) shall be subjected to the rough handling tests in accordance with Specification MIL–P–116, applicable only to level A.

5. PREPARATION FOR DELIVERY

5.1 Materials and methods. The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

5.1.1 *Materials.* The following shall apply wherever packaging materials are referred to in this specification by name only:

Reference	*Full identification*
Gummed tape	Class 2 of Specification UU–T–111.
Paperboard cushioning	Type I, class 1 of Specification PPP–P–291.
Water-vaporproof bag	Type I, class e of Specification MIL–B–117.

5.2 Size of containers. The bag, interior and exterior containers shall be of a minimum size that will adequately contain the items.

5.3 Packaging.

5.3.1 *Level A.*

5.3.1.1 *Cleaning.* Cleaning Process C–1 of Specification MIL–P-116 shall be followed to clean the compass and the carrying case.

5.3.1.2 *Drying.* Immediately after cleaning, items shall be thoroughly dried in accordance with any applicable procedure specified in Specification MIL–P–116 that will not damage the items.

5.3.1.3 *Unit packaging.* Packaging shall be as specified herein, which is in conformance with Method IA–8 of Specification MIL–P–116.

5.3.1.3.1 *Publication TM9–6605–200–12T and compass without carrying case.*

5.3.1.3.1.1 *Wrapping.* Publication TM9–6605–200–12T shall be folded and placed with the compass. The compass and publication shall then be wrapped with paperboard cushioning as illustrated in Figure 1, except that the publication is not shown with the compass. The wrap shall be secured in place with gummed tape.

5.3.1.3.1.2 *Water-vaporproof bag.* The wrapped compass and publication shall be placed in a water-vaporproof bag. The bag shall be heat sealed in such a manner as to pass the heat seal and quick leak tests of Specification MIL–P–116. Air shall be exhausted from the bag prior to final closure.

5.3.1.3.1.3 *Paperboard box.* Each bagged compass and publication shall be immobilized in a paper board box conforming to type I or IV, any class and any style of Specification PPP–B–676. The box shall be closed as specified in the appendix of Specification PPP–B–676.

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5.3.1.3.2 *Publication TM9–6605–200–12T and compass with carrying case.*

5.3.1.3.2.1 *Wrapping the items.* The compass shall be placed in its carrying case and the cover shall be secured. Publication TM9–6605–200–12T shall be folded and placed with the case containing the compass. The publication and the case shall then be wrapped with paperboard cushioning as illustrated in figure 2, except that the publication is not shown with the case.

5.3.1.3.2.2 *Water-vaporproof bag and paperboard box.* The wrapped items shall be placed in a water-vaporproof bag. The bag shall be heat sealed and air exhausted as specified in 5.3.1.3.1.2. The bagged items shall be immobilized in a box conforming to the characteristics (except for size) of the box specified in 5.3.1.3.1.3. The box shall be closed as specified in 5.3.1.3.1.3.

5.3.1.4 *Intermediate packaging.* Unless otherwise specified, only items with the same stock number shall be placed in an intermediate container. Then paperboard boxes, specified in 5.3.1.3.1.3 or 5.3.1.3.2.2, shall be immobilized in a fiberboard box conforming to class 2, W6c, any compliance symbol and any style, except FTC, TS, OPF, and FPF, of Specification PPP–B–636. The fiberboard box shall be closed with two strips of 2-inch width pressure-sensitive tape conforming to Specification PPP–T–76. The tape shall be applied to the top and bottom seams and extend at least 3 inches onto each end panel.

5.3.2 *Level C.*

5.3.2.1 *Cleaning and drying.* Cleaning and drying shall be as specified in 5.3.1.1 and 5.3.1.2.

5.3.2.2 *Unit packaging.* Packaging shall be as specified herein, which is in conformance with method III of Specification MIL–P–116.

5.3.2.2.1 *Wrapping and boxing compass without a carrying case.* Wrapping shall be as specified in 5.3.1.3.1.1. Each wrapped compass or a compass with a publication shall be immobilized in a paperboard box conforming to any style, except style VI, any type or class of Specification PPP–B–566. The paperboard box shall be closed as specified in the appendix of Specification PPP–B–566.

5.3.2.2.2 *Compass with carrying case.* The compass shall be placed in its carrying case and the cover shall be secured. A folded publication TM9–6605–200–12T and the case containing a compass shall be immobilized in a box conforming to the characteristics (except for size) of the box specified in 5.3.2.2.1. he box shall be closed as specified in 5.3.2.2.1.

5.3.2.3 *Intermediate packaging.* Unless otherwise specified, only items with the same stock number shall be placed in an intermediate container. A number of paperboard boxes shall be immobilized in a fiberboard box conforming to type I or II, class 1, any style, except FTC, TS, OPF and FPF of Specification PPP–B–636. The box shall be closed as specified in 5.3.1.4, except that the 2 inch width tape shall conform to gummed tape. The gross weight of the intermediate package shall not exceed twenty pounds.

5.4 **Packing.** Unless otherwise specified, all the items in the intermediate packages packed in a shipping container shall contain the same stock number.

5.4.1 *Level A.* Twelve intermediate containers shall be immobilized in a shipping container conforming to one of the following specifications:

PPP–B–601 — Style A or B (oversea type)—type II or III, class 2 plywood.

PPP–B–621 — class 2 (oversea), style 2 or 4.

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5.4.2 *Level B.* Twelve intermediate containers shall be immobilized in a shipping container conforming to one of the following specifications:

PPP–B–585 — Any style except style 1, class 1, construct for type 3 load.

PPP–B–601 — Any style under domestic type.

PPP–B–621 — Class 1 (domestic), style 2 or 4.

5.4.3 *Level C.* A quantity of intermediate containers containing unit packaged compasses, with or without carrying cases, shall be immobilized in a shipping container conforming to type I or II, class 1, any style, except FTC, TS, OPF and FPF of Specification PPP–B–636. The gross weight shall not exceed the weight limits specified in Specification PPP–B–636.

5.5 Marking.

5.5.1 *Interior packages.* Each bag and each box shall be marked in accordance with Standard MIL–STD–129. The serial number shall be added in the identification data on each level A unit package and serial numbers shall be indicated in the identification data on each level A intermediate container. Fragile markings shall be applied as specified in Standard MIL–STD–129.

5.5.2 *Shipping containers.* Each shipping container shall be marked in accordance with Standard MIL–STD–129. If the serial numbers of the compasses in a level A or B shipping container are in consecutive order, the serial numbers shall be indicated as specified in Standard MIL–STD–129. If the serial numbers are not in consecutive order, they shall be indicated on packing lists. Fragile markings shall be applied as specified in Standard MIL–STD–129.

5.5.3 *Stock number.* The stock number to be added in the identification data shall be obtained from the contracting officer.

5.6 Preproduction pack. Prior to quantity packaging production, a preproduction pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification. A rough handling test is required.

5.7 Workmanship. All operations involved in accomplishing corrosion, prevention, preservation, packaging and packing by the procedures specified herein shall be in accordance with the highest grades of practice associated with this type of work.

6. NOTES

6.1 Intended use. The Compass: M2 is a multipurpose instrument used to obtain angle of sight and azimuth readings. The limits of operation: in azimuth, 0 to 6,400 mils; in elevation, 0 to plus or minus 1200 mils.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).
(e) The quantity of Publication TM9–6605–200–12T to be provided for each contract on the basis of 1 publication for each compass.

6.3 Design information. Design information regarding specific equipment not covered or referenced by this specification may be obtained from the preparing activity upon the specific request of the contractor.

6.4 Miscellaneous notes.

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6.4.1 Special precautions are recommended to insure that purchased materials, including organic and inorganic finishes are free from contamination by iron and other magnetic materials which might have an adverse effect upon the accuracy and reliability of the magnetic compass.

6.5 Size and weight of packed shipping container.

6.5.1 *Compass without carrying case.* The following inside dimensions, cubage and gross weight are approximately for twelve intermediate packaged Compasses: M2 without carrying cases, specified in 5.3.1.4, packed in a nailed wood box for Level A:

Inside dimensions	— 24. inches long by 10.5 inches wide by 8. inches deep.
Cubage	— 1.7 cubic feet.
Gross weight	— 115. pounds.

6.5.2 *Compass with carrying case.* The following inside dimensions, cubage and gross weight are approximately for twelve intermediate packaged Compasses: M2, with carrying cases, specified in 5.3.1.4, packed in a nailed wood box for level A:

Inside dimensions	— 32. inches long by 24. inches wide by 12.5 inches deep.
Cubage	— 7.2 cubic feet.
Gross weight	— 175. pounds.

6.6 Definitions.

6.6.1 *Gage.* A gage includes all precision standards and appilcable accessories which are calibrated to one basic size to provide an accurate reference dimension.

6.6.2 *Device, special testing (DST).* Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.6.3 *Standard measuring equipment (SME).* Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, and so forth.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligations whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodians:
Army—Ord
Navy—BuShips
Air Force—MAMA

Preparing activity:
Army—Ord

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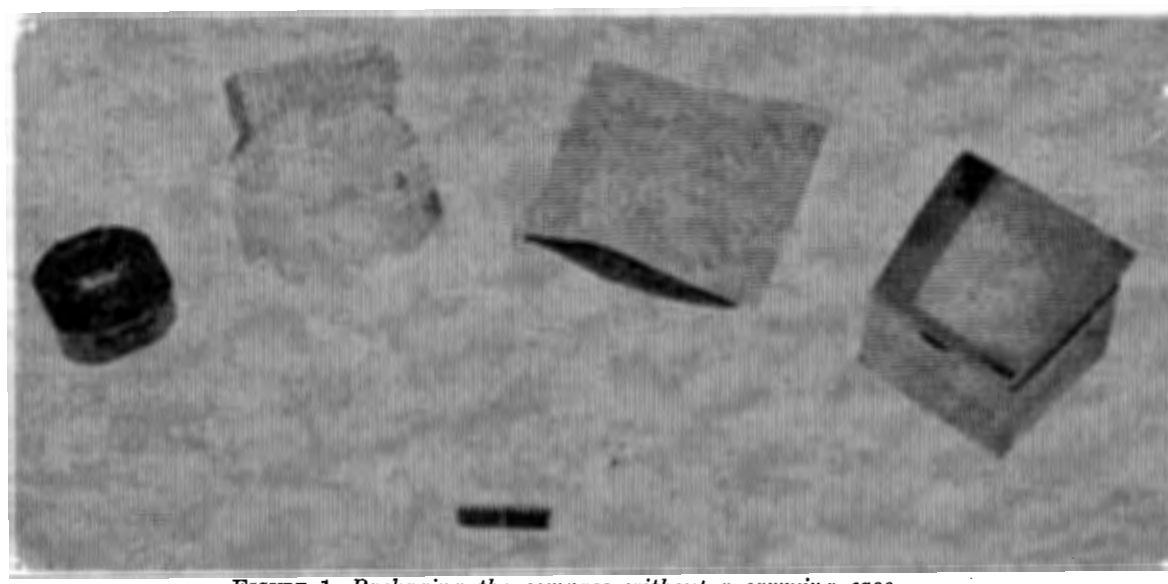


FIGURE 1. *Packaging the compass without a carrying case.*

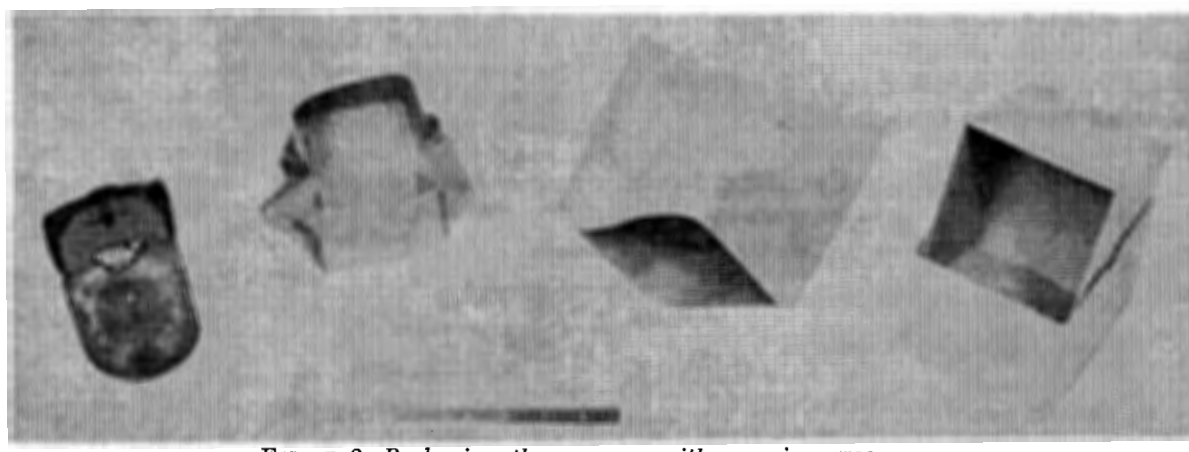


FIGURE 2. *Packaging the compass with carrying case.*

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